

Barchart Histogram

Matplotlib tasks

Task 1 Horizontal Bar Chart

- Create a **horizontal bar chart** for:
 - Categories: Tea, Coffee, Juice, Water
 - Quantity: any 4 numbers
 - Change:
 - Bar color
 - Title
 - X-label and Y-label
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Task 2 Vertical Bar Chart

- Create a **vertical bar chart** showing:
 - Students: A, B, C, D
 - Marks: random values between 40–100
 - Set:
 - width = 0.4
 - edgecolor = black
 - alpha = 0.8
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Task 3 Histogram (Basic)

- Generate **50 random numbers** between 1–50
 - Plot a histogram
 - Add:
 - X label
 - Y label
 - Title
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□ MEDIUM TASKS

Task 4 Bar Chart with Different Colors

- Create a bar chart for:
 - Fruits: Apple, Mango, Banana, Orange
 - Sales: [20, 40, 35, 60]
 - Use **different colors for each bar**
 - Add `edgecolor='black'`
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Task 5 Horizontal Bar Chart (Stock Data)

- Products: Pen, Pencil, Eraser, Scale
 - Stock values: random numbers
 - Plot **barh**
 - Change:
 - Transparency
 - Title
-

Task 6 Histogram with More Data

- Generate **100 random values**
 - Plot histogram
 - Change:
 - Number of bins
 - Title text
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● THINKING TASKS (Still Easy)

Task 7 Compare Two Charts

- Plot:
 - One **vertical bar chart**
 - One **horizontal bar chart**
 - Use **same data**
 - Observe: Which looks clearer?
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Task 8 – Histogram Meaning

- Generate random marks of **100 students**
 - Plot histogram
 - Answer (in comments):
 - Which marks range has the highest frequency?
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Task 9 – Real-Life Data

- Create a bar chart for:
 - Days: Mon, Tue, Wed, Thu, Fri
 - Study hours per day
 - Add:
 - Title
 - Labels
 - Colors
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🔥 MINI CHALLENGE (Optional)

Task 10 – Fruits Report

- Create:
 1. **Bar chart** → Fruit quantities
 2. **Histogram** → Random customer purchase counts
- Both in **separate plots**